

Local Floer Chain Complex across Hamiltonian Bifurcation

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Contents

This talk is based on the ongoing work with Hong-kwon Jo (SNU).

- **Motivation and Preliminaries.**

Bifurcation, Conley–Zehnder index and Floer homology.

- **Classification of Hamiltonian bifurcations.**

Birth-death, period doubling, phantom kiss and emission.

- **Chain complex mutation.**

Construction of cylinders, Morse–Bott cascade counting.

Main result. Describing the explicit change of Conley–Zehnder indices and Floer chain complex for generic Hamiltonian bifurcation types.

Motivation

Motivation 1. Bifurcations

A **bifurcation** is, heuristically, a birth and death of periodic orbits.

Recently, there are many works [AFvK⁺23], [AB25], [JKvK26] relating bifurcation with symplectic invariants, for example Conley–Zehnder indices, Floer-type Euler characteristics, etc.

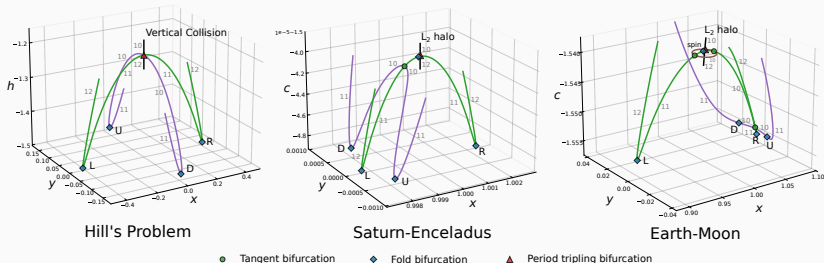


Figure 1: Bifurcation in Hill's lunar problem and CRTBP, from [JKvK26].

Classical work of Meyer [Mey70] classified generic bifurcation types. Can we make a *symplectic dictionary* to these bifurcation types?

Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

The multiple covers of the retrograde and direct orbits bifurcates, which produces orbits looks like a flower.

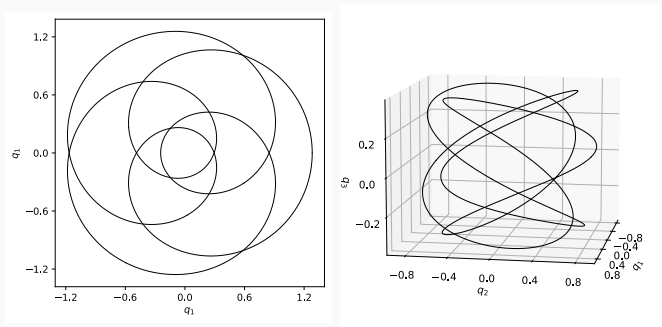
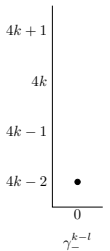
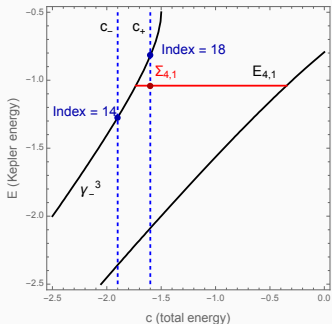


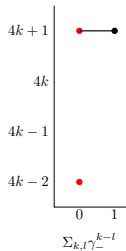
Figure 2: Resonant orbits of RKP, in courtesy of Chankyu Joung.

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed this bifurcation can be translated to a change of Floer chain complex in terms of spectral sequences.



At $c_{k,l}^- - \varepsilon$



At $c_{k,l}^- + \varepsilon$

Slogan. Bifurcation gives a *mutation of a Floer chain complex*.

We aim to describe this effect explicitly.

Preliminaries

Bifurcation

Let $H_\varepsilon : W^4 \rightarrow \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0 , an eigenvalue of the linearized return map on the Poincaré section.

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along bifurcation parameter.

If $\lambda_0 = 1$, so the orbit degenerates, our orbit cylinder may break:

- periodic orbits may be born or vanish, called **bifurcation**,
- if $\lambda \neq 1$ but $\lambda^k = 1$, then the k -th cover bifurcates.

Example. Period Doubling

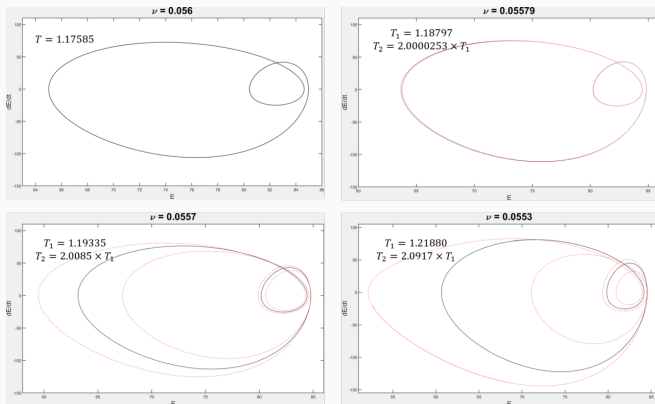


Figure 3: Period doubling in Kuramoto–Sivashinsky equation, from [Max21].

Period doubling is a generic bifurcation of double cover.

The red-dashed orbit is the new orbit emerged from the bifurcation.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{CZ}(\gamma)$ is an integer associated to a periodic orbit, might be understood as a *symplectic winding number*.

Our interest is a change of index along the parameter, $\mu_{CZ}(\gamma_\varepsilon)$.

Along a non-degenerate family, the CZ index is constant.

Key idea. Change of μ_{CZ} = Degeneracy of γ_ε = Bifurcation occurs

Hence, a bifurcation can be detected via change of the CZ index.

(cf. [AB25], [JKvK26], etc.).

Floer Homology

A **Floer homology** $HF(W, H)$ is a topological invariant associated to a symplectic manifold W and Hamiltonian H . (cf. [AD14].)

This is determined by **Floer chain complex** $CF(W, H)$, whose

- generators are 1-periodic orbits of H ,
- grading is given by CZ indices,
- differential is determined by a count of **Floer cylinder**, whose boundaries are periodic orbits.

Through a bifurcation, $CF(W, H)$ changes a lot;

- orbits are born or vanish $\Rightarrow$ number of generators changes.
- CZ index changes $\Rightarrow$ grading changes

However, the homology doesn't change, so the differential is in charge to make the homology consistent. *Can we describe this explicitly?*

Classification of Bifurcations

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling, murder (opposite stability type)
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Aside. We will use names from [AM78].

There are 3 extra types given, which include critical points.

Other names: touch-and-go, period tripling/quadrupling, etc.

Birkhoff Normal Form

We find the periodic points of the iterated return map Φ_ε^k .

This can also be done in essentially same way, by finding critical points of the generating Hamiltonian of the return maps.

1. We can write the generating Hamiltonian of the return map Φ by

$$G_t = G_2 + \sum_{k \geq 3} G_{k,t}$$

where $\deg G_k = k$, $G_2^{ss} = \alpha(u^2 + v^2)$, $\dot{G}_{k,t} = \{G_2^{ss}, G_{k,t}\}$, $\lambda = e^{i\alpha}$.

2. If $\lambda_0^k = 1$, the generating Hamiltonian of k -th iterate of the return map Φ^k can be chosen with $G_2^{ss} = 0$, so $\dot{G}_{k,t} = 0$.
3. This is map on a *covering space* of the neighborhood of γ , which should descend to the base space to generate the first return map, which forces a symmetry of G .

Generating Hamiltonians – Birth-death

Generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 - \varepsilon u + \dots$$

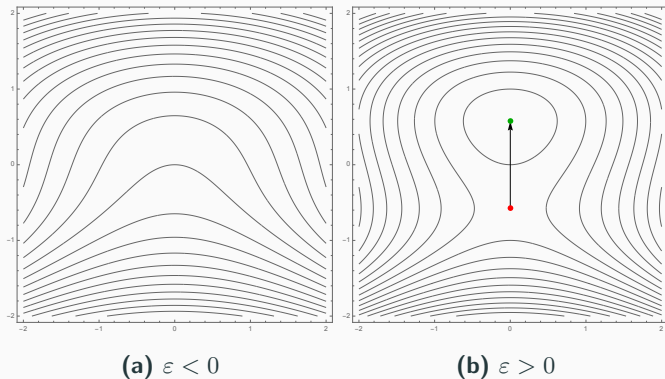


Figure 4: Critical points and gradient trajectories of birth-death bifurcation

Generating Hamiltonians – Period doubling

Generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 - \varepsilon u + \dots$$

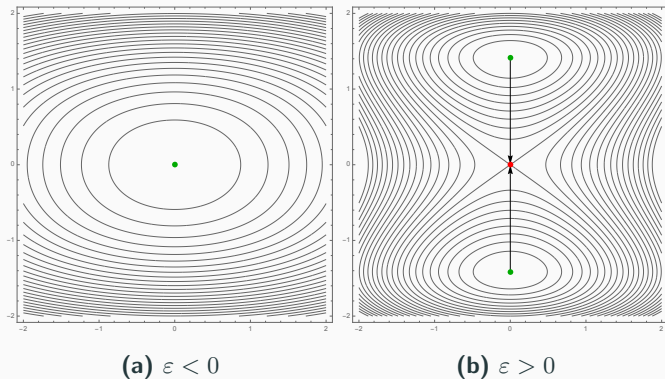


Figure 5: Period doubling bifurcation

Generating Hamiltonians – Phantom kiss

Generic 3rd or 4th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^k \cos k\theta - \varepsilon r^2 + \dots$$

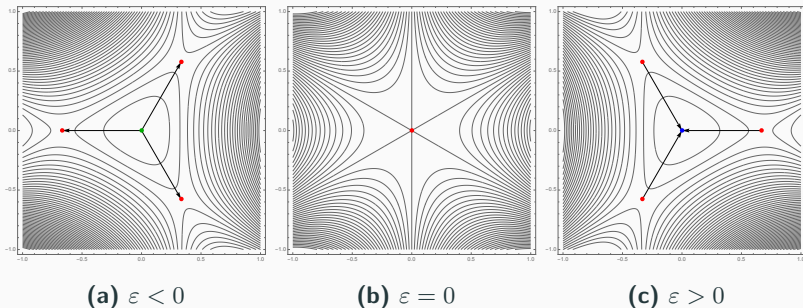


Figure 6: Phantom kiss bifurcation

Generating Hamiltonians – Emission

Generic $k \geq 4$ -th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^4 + \beta r^k \cos k\theta - \varepsilon r^2.$$

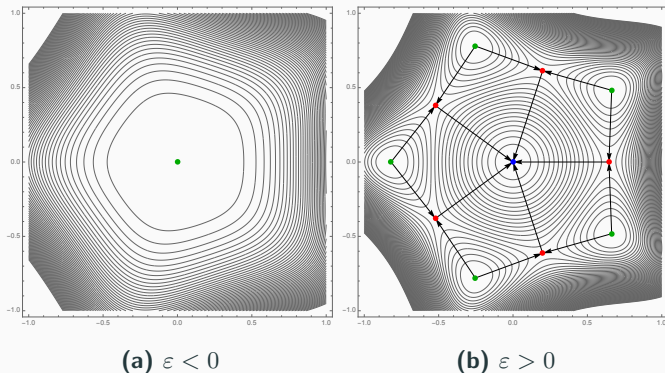


Figure 7: Emission bifurcation

Note. 4th cover is a borderline case, depending on $|\alpha|$ and $|\beta|$.

Summarization

Name	Floquet Mult.	Before	Stability	Morse Index	After	Stability	Morse Index
Birth-death	$\lambda = 1$				γ_E	Elliptic	0 or 2
					γ_H	Hyperbolic	1
Period doubling	$\lambda = -1$	γ^2	Elliptic	0 or 2	γ^2	Hyperbolic	1
					γ_E	Elliptic	0 or 2
Murder	$\lambda = -1$	γ^2	Hyperbolic	1	γ^2	Elliptic	0 or 2
		γ_E	Elliptic	0 or 2			
Phantom Kiss	$\lambda^k = 1$ $k = 3, 4$	γ^k	Elliptic	0 or 2	γ^k	Elliptic	2 or 0
		γ_H	Hyperbolic	1	γ_H	Hyperbolic	1
Emission	$\lambda^k = 1$ $k \geq 4$	γ^k	Elliptic	0 or 2	γ^k	Elliptic	2 or 0
					γ_H	Hyperbolic	1
					γ_E	Elliptic	0 or 2

Table 1: Generic Hamiltonian bifurcations

Note. There are more bifurcations, e.g. pitchfork, which require extra symmetry of the system or the orbit.

Chain Complex Mutation

Easy example. Birth-death bifurcation

Let an orbit γ_0 experiences the birth-death bifurcation along ε .

1. If $\varepsilon < 0$, there is no orbit.
2. If $\varepsilon > 0$, there are two orbits, say γ_H and γ_E .

(Very simplified) **Floer-theoretic viewpoint.**

1. If $\varepsilon < 0$, $CF = 0$.
2. If $\varepsilon > 0$, $CF = \langle \gamma_H, \gamma_E \rangle$. But both produces same homology.

Hence, $|\mu_{CZ}(\gamma_H) - \mu_{CZ}(\gamma_E)| = 1$ and they should cancel each other, so there exists a Floer cylinder between them. (cf. [GG10].)

However, the situations are more complicated for higher covers.

Main Result and Key Ideas

Main result. Full description of the mutation of Floer chain complex through 5 generic bifurcation types.

Key Ideas.

1. Construct a Hamiltonian H_ε which represents some bifurcation type via adiabatic approximation.
2. Gradient trajectories of G_ε gives Floer cylinders of H_ε .
3. We get cylinders connecting a simple orbit and a multiple cover.
 $\Rightarrow$ A contribution of 1 cylinder may not be exactly 1.
Using Morse–Bott cascade, we can compute the explicit chain complex, of which the resulting homology is invariant.

Adiabatic approximation of Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian of k -th iterate of the return map.

Using **adiabatic approximation**, which is an integration along the orbit, we construct the model Hamiltonian for each bifurcation type,

$$H_\varepsilon = h(r) + G_\varepsilon.$$

- H_ε is defined on the covering space, but Floer-type informations can be descended to the base space.
- H_ε is not exactly same as $\tilde{H}_\varepsilon$, but has the same return map and Conley–Zehnder indices with $\tilde{H}_\varepsilon$.

Morse–Bott Cascade

Our periodic orbits have an S^1 -symmetry by the time shift, which makes the situation slightly different from the usual Floer theory.

We equip a Morse function on each orbit, and consider two critical points (maximum and minimum) as generators of a Floer chain complex.

In this case, differential is given by counting **Morse–Bott cascade**, a sequence of Morse trajectories and Floer cylinders.

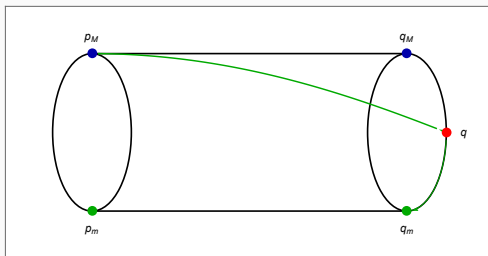


Figure 8: Morse–Bott cascade.

Chain complex mutation of Period doubling

1. If $\varepsilon < 0$, there is one orbit γ .
2. If $\varepsilon > 0$, there are two orbits; γ^2 and γ_H .
3. $\mu_{CZ}(\gamma^2)$ changes, say N to $N - 1$.
4. $\mu_{CZ}(\gamma_H)$ must be N , and there exists 1 Floer cylinder between γ^2 and γ_H , whose contribution is given as following.

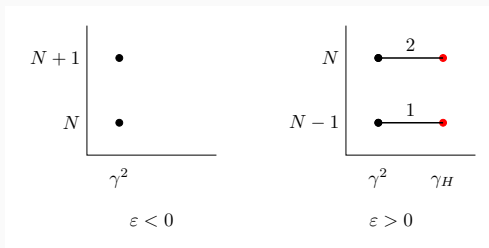


Figure 9: Chain complex mutation across the period doubling.

$\gamma_{H,\min}$ and $\gamma_{\min}^2$ cancel each other, while $\gamma_{H,\max}$ and $\gamma_{\max}^2$ don't.

Chain complex mutation of Phantom kiss and Emission

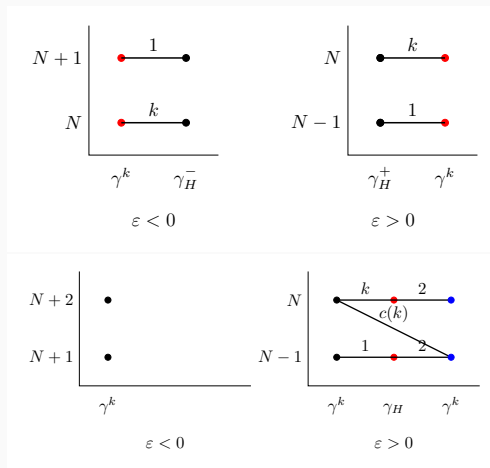


Figure 10: Chain complex mutations across phantom kiss and emission



Figure 11: Thank you for your attention!

Questions, comments, and further discussions are welcomed!



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Illustration of period doubling in the Kuramoto–Sivashinsky equation. CC0 1.0 Universal Public Domain Dedication.



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